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Utilizing Historic Borehole Acoustic Data for Sub-Seismic Imaging and Advanced Reservoir Properties

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Abstract

Existing historic full waveform borehole acoustic data can be utilized to obtain formation information previously overlooked or unavailable, due to new processing methods or updated reservoir management requirements, without the need for new data acquisition.

While seismic data is often reprocessed due to new developments in processing algorithms, full waveform borehole acoustic data from array acoustic tools are often overlooked as a source of additional reservoir information. Full waveform monopole and dipole waveforms contain a wealth of information beyond the basic compressional and shear slowness results typically provided. Slowness resolution enhancement, stoneley permeability profile calculation, azimuthal anisotropy, transverse anisotropy, and deep shear wave imaging are all viable additional processing outputs from existing datasets.

Multiple additional results can be extracted from existing data, without drilling a new well, or in order to optimize drilling or logging programs for new wells. Resolution enhancement of compressional and shear slowness may be used to better delineate thin bedded environments. Stoneley Permeability provides a continuous permeability profile to compliment known mobility points from formation testing, or to provide a baseline to assess the long term production performance. Anisotropy processing gives valuable input into the geomechanical setting, and can enhance seismic processing by fine tuning the velocity field into three dimensions. Deep Shear Wave Imaging can be performed on existing cross-dipole datasets to give sub-seismic imaging of acoustically reflective surfaces such as bed boundaries or faults more than 50ft away from the wellbore, increasing the reach of the acoustic logs far from the wellbore.

Reprocessing of historically recorded full waveform borehole acoustic data is a cost effective way to obtain additional formation information to enhance reservoir models or plan future field developments.

Introduction

Logging of open hole wells provides a unique opportunity to acquire the best possible quality formation data. After a well has been completed with steel casing, many logging technologies are no longer usable, or their quality is reduced by the presence of casing or quality of the cement. As such, the appropriate planning of the initial open hole logging program to acquire all necessary data is of great importance. Geoscientists would generally prefer a comprehensive program of advanced logging for almost all sections and wells to

constrain all the nuances and complexities of the intersected geology, however real-world budget constraints limit what is practical in all but the most vital wells.

In some cases, unexpected challenges or complexities may be uncovered later in the lifecycle of a well, which would have changed the initial open hole logging program had the need been anticipated in the well planning and construction phase. If additional infill or offset wells are planned in the vicinity of the geological concern it may be possible to acquire a more complete logging suite to further assess the problem, however this is rarely the case. If appropriate logging through casing measurements are not possible in the original well, either due to completion or tool physics limitations, the reassessment of on any existing original well data is the only viable option.

Similarity to Seismic Data Reprocessing

The concept of reprocessing historic data is certainly not a new concept in the industry, with old seismic data routinely reprocessed when new surveys are conducted in the same area. By reprocessing the old seismic data it is possible to use similar processing parameters to the new data for over consistency of the final seismic images. Seismic tie points from new wellbores, or other depth matching to new seismic acquisition can be standardized across the survey area; or updated velocity fields may be applied. If any new processing techniques have been made available since the original processing, it may be possible to enhance the old data for new insights into the geology of the area.

Seismic reprocessing of this type is possible due the existence of original raw seismic data, and this multitude of raw information can be handled in any way that is deemed appropriate by the seismic analyst. Similarly, certain wireline or Logging while drilling (LWD) tools also record large volumes of raw data that is processed for the typical final output curves. Common examples of devices that record large volumes of raw information for later processing are Nuclear magnetic resonance (NMR), borehole imaging logs, and acoustic logs. When the raw data from these tools is available, it is possible to either reprocess the data in a similar manner to the original processing with perhaps a selection of different parameters, or in a new way not considered at the original time of logging. Data may be in an industry standard format, or a proprietary format of the logging service company, which influences the available options for reprocessing either by an external commercial package, or with the original logging company software and systems.

Many modern logging tools can be run in different acquisition modes to record different levels of data to suit a customer's requirements. In the case of acoustic wireline tools, this may range from basic first arrival slowness measurements, through fullwave monopole and inline dipole modes, and end with full cross-dipole logging or other specialized acquisition modes. When only lower-level logging requirements have been requested, for instance just for compressional and shear slowness values, it is possible that additional data up to cross-dipole data may have also been recorded. This extra data may be recorded by the logging company for the sake of simplicity by using a single default acquisition mode, as the full data set can be logged by modern tools without any reduction in logging speed or data quality. If this additional data has been recorded, it can be possible, by mutual agreement, to upgrade at a later date to more advanced processing products. Raw data of this type may be available from wells logged many years previously, assuming it has been archived and is retrievable.

Challenges faced when retrieving historic data

Retrieving old data may not be a trivial exercise, depending on a number of factors including the nature of the archive system, digital storage media and the hardware required to read such media, old digital data formats, and auxiliary data requirements to undertake processing. These topics are discussed below.

Archive system

Logging contracts or local government regulations typically have a minimum retention time for service companies to retain all logging data. During this mandated time period the data will be appropriately stored by the logging contractor and would be available for subsequent reprocessing. At the end of this period, or potentially at the end of a logging contract, the availability of data may decrease either due to required destruction or deletion of the data, or else the archive not being maintained after the minimum period.

Hardware

Before the widespread availability of large capacity hard-disk drives, the large volumes of logging data from acoustic full-wave logging of image logs were often stored on digital tape cartridges, likely transitioning after this period to storing data on DVD media, and more recent logging jobs being archived directly onto hard-disk systems. The hardware to download from these earlier tapes is generally now out of use and finding active and useable tape drives may be an added complexity in the quest to retrieve and reprocess older data. It is likely that in the not distant future the ability to retrieve data from older formats may be very limited or become a service only available from specialist data retrieval companies at significant cost.

Media degradation

All digital media will degrade over time, a process that will accelerate if they are not stored in appropriate conditions. It is possible that recovered digital tapes or DVD's that have been archived or otherwise in storage, even with the available hardware to retrieve the data, may already be unreadable.

Data formats

The evolution of data formats in all digital realms continues to progress at a high rate, which even a casual computer user may encounter when trying to access older files. Logging company proprietary formats tend to be long lived and are often backwards compatible across the timeframe of the logging tools being considered. This longevity is to the advantage of a user wishing to extract and reuse historical datasets. Industry standard logging formats have also evolved over time to enable more complex datasets to be stored, and once again the general stability and backwards compatibility of these standard formats is an advantage for a potential reprocessing request.

Auxiliary data

With the successful archive retrieval of acoustic waveform data, it is still possible that other auxiliary data required for processing, such as orientation data, is still missing. Maintaining clear data requirements for different processing solutions is an important step in this process. Keeping clear guidelines for required deliverable data is also important for current logging jobs, to ensure the appropriate data is available for any future reprocessing efforts.

Acoustic Data Processing options

When old full-wave acoustic data has been made available, there are any number of processing options available depending on the geological requirements or problems that need additional information. At the most a basic level, original data could be reprocessed for acoustic slowness. Slowness reprocessing could be initiated due to low confidence in the initial results, perhaps due to an automated slowness picking process originally used, or due to complex geological responses that were not fully understood at the time of initial logging. Incorporating later acquired formation knowledge during the reprocessing can resolve the initial uncertainties. The historic logging contract may have only specified the delivery of compressional data, and shear or stoneley slowness is now required.

In the following sections, some additional processing options will be examined that could add appropriate formation information for different geological situations. The highlighted topics are not a complete list of the options available from acoustic data but do cover a wide range of common challenges.

High Resolution Processing

The standard resolution of acoustic fullwave processed data using semblance or correlation methods is controlled by the size of the receiver array. For an eight-receiver tool with Transmitter to Transmitter spacing of 0.5 feet, the receiver array covers a distance of 3.5ft. If layers in the formations are smaller than this interval, the resultant slowness values will be some combination or average of the individual layer values, potentially masking good reservoir zones. It is possible to increase the resolution of the output slowness by reducing the number of receiver wave-traces used, but this will tend to increase the noise levels as overall less data is used in the correlation. An alternate technique using waveform matching/inversion method (Zhang et al, 2000) and utilizing overlapping subarrays across the same depth interval. The larger numbers of overlapping arrays used in this method does not suffer the same increase in noise as encountered when simply reducing the number of levels used in the correlation. The overlapping receivers is the result of the logging increment (typically 0.5 ft.) being equal to the inter-receiver spacing of the receiver array. Different resolution options are possible utilizing a receiver aperture between 0.5 and 3.5ft (Figure 1). The enhanced resolution compressional or shear slowness from existing monopole or shear waveform data allows for a more accurate formation evaluation and reserve estimation in thin bedded environments. Figure 2 shows the improved level of detail available for compressional slowness in a thinly bedded environment when using smaller apertures.

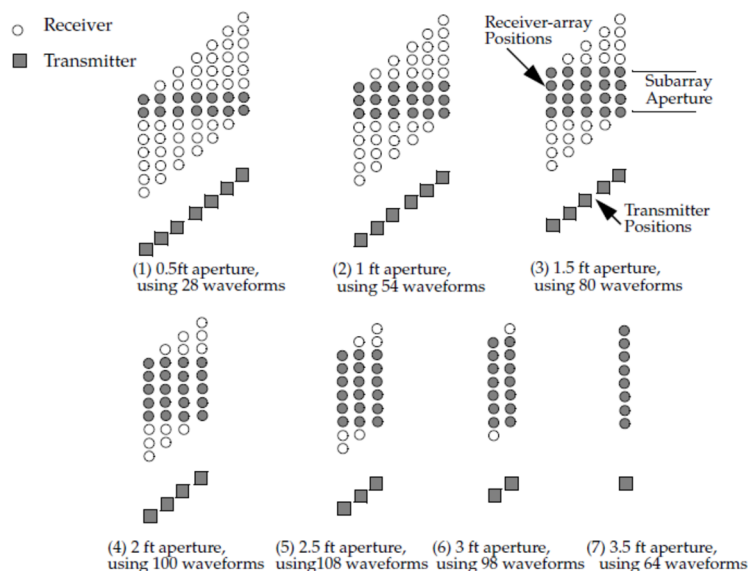


Figure 1—Configuration of receiver wave-trace data as used for enhanced resolution slowness processing. Variable receiver aperture lengths from 0.5ft to standard 3.5ft may be selected to better delineate thin bedded sequences.

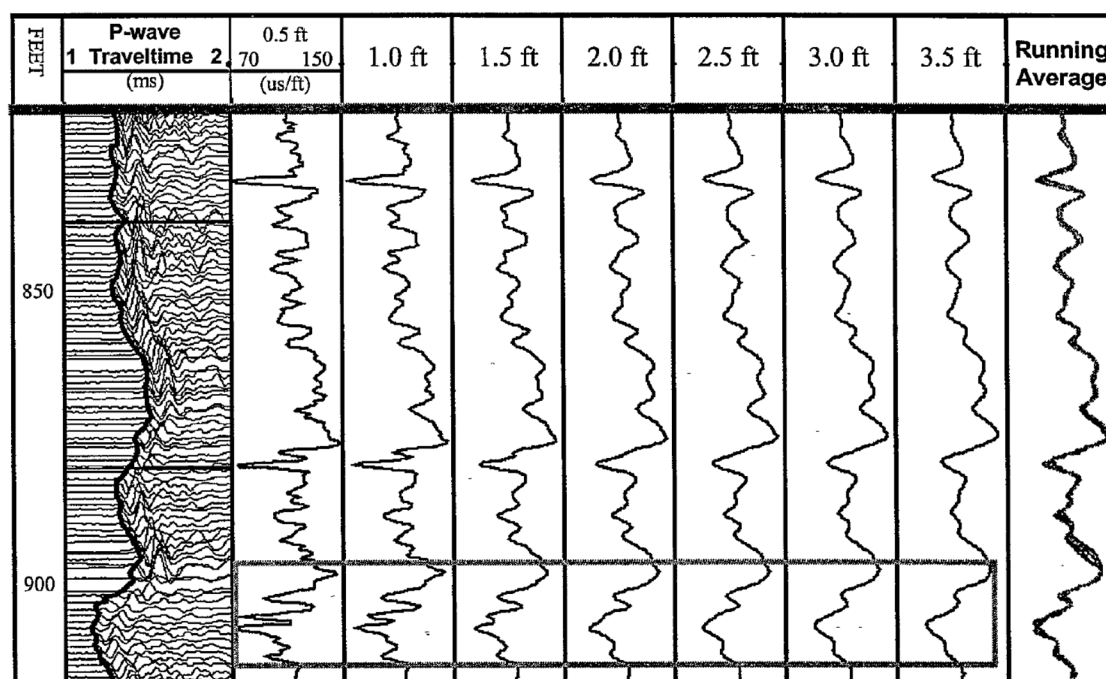


Figure 2—Example of compressional slowness response from receiver aperture lengths from 0.5ft to standard 3.5ft, illustrating the higher level of detail possible in thin bedded sequences (Zhang et al, 2000)

Stoneley Permeability

Knowledge of the permeability of formation is an essential property needed to plan and manage the extraction of hydrocarbons from a reservoir. Well testing or downhole formation testing tools yield permeability (or mobility) data at different scales, however neither method provides a continuous permeability profile to indicate the variability changing with depth. Stoneley waves are sensitive to permeability, where the permeability effect can significantly increase the Stoneley slowness and cause its frequency content to shift to lower frequency, and when a formation is permeable, the hydraulic exchange between borehole and formation changes the Stoneley wave propagation characteristics (Tang and Patterson, 2004). These changes in the Stoneley wave velocity and attenuation properties are used to determine a continuous permeability profile, filling the gaps between formation testing points and adding additional detail missing from larger scale well testing.

Stoneley wave data required for this type of permeability processing are found in the low frequency content of fullwave monopole waveforms, or in a specific low frequency acquisition sequence for some logging tools. Stoneley permeability processing may also be undertaken on monopole data from Logging While Drilling (LWD) acoustic tools. As discussed previously, these waveform data are commonly recorded and may be available for later analysis.

Stoneley permeability processing may be utilized for newly drilled and logged wells, but it could equally be performed on historic acoustic data where the need for this analysis was either not identified or required at the time. For instance, it is typical for older production wells to experience a reduction in performance, at which time a properly planned intervention may improve the well productivity. Conducting production logging surveys is a useful way to determine the contribution from different formations or intervals, however for cased and perforated completions the flow information is limited to those zones in communication with the formation via the perforations. When additional knowledge of the formation from behind the casing is required to determine the best way to improve well performance or update a reservoir model, the use of original openhole logs is the only available choice. The additional of the original permeability profile from previously acquired acoustic data may assist in better assessing the flow characteristics of the well, and to aid in the formulation of an intervention plan.

Anisotropy Analysis

Anisotropic formation properties have a multitude of influences on oilfield processes and well construction tasks. Anisotropy may be intrinsic to the formation as the result of depositional processes such as layering, or reflective of stress or stress-induced changes due to current or previous geomechanical conditions. The understanding of these effects has a direct impact on, among other things, the choice of stable drilling azimuths, hydraulic fracture performance, the depth or location accuracy of surfaces in seismic volumes, or drainage volumes influenced by connected fracture systems. While gathering information for these purposes may be primary objective of a well data acquisition program, it is also possible that the need for anisotropy data is only recognized after drilling difficulties or other complications are encountered. Alternatively, reprocessing old data from a nearby well may prove useful to successfully plan a new well.

Prior to the introduction of dipole transmitters, processing techniques were developed to derive shear-wave slowness from stoneley-wave data (Stevens and Day, 1986). While dipole tools have replaced the stoneley method for shear slowness determination, the relationship between stoneley and shear can still serve a purpose, particularly as the stoneley wave is controlled by horizontal rather than vertical shear velocity (Tang and Patterson, 2004), assuming a vertical wellbore. Comparing the vertical and horizontal shear values gives a measure of transverse anisotropy. Layered formations such as shales or thin interbedded layer often display intrinsic transverse anisotropy, and the failure to consider this variability may adversely affect seismic migrations or amplitude versus offset (AVO) analyses. An extension of the process used to determine stoneley permeability (as discussed in the previous section) can be used to extract the horizontal shear slowness. The primary requirement for this processing is the fullwave monopole data, although a vertical shear measurement is required for comparison.

Azimuthal anisotropy processing requires full cross-dipole data and tool orientation data in order to resolve the anisotropy magnitude and direction. The directional component of flexural waves will enable the detection and measurement of shear wave splitting. The fast-shear and slow-shear components are extracted from the full cross-dipole data using mathematical or inversion methods (Alford, 1986; Tang and Chunduru, 1999). Azimuthal anisotropy measures the difference in flexural shear slowness at different directions around the wellbore at different azimuths, so for vertical wells this direction equates to variations matching a given compass direction. In contrast, when recording data in horizontal wells, the resultant fast and slow directions may align with horizontal and vertical slowness responses. The orientation of the well relative to layering or stress field will determine the information that can be extracted and should be considered when interpreting this data. The combination of the azimuthal anisotropy and transverse anisotropy data allows the definition of the 3 dimensional velocity field near the wellbore, which can be extrapolated and incorporated into larger 3D seismic surveys.

Deep Shear Wave Imaging

Most wellbore acoustic applications or outputs consider vibrational energy or wave behaviour in the immediate vicinity of the borehole, with a depth of investigation up to a maximum of around 3 feet. While acoustic logging tools are designed to optimise slowness measurements in a variety of logging conditions, the idea of using the same tools and recorded waveforms to isolate reflected signals from deeper in the formation has long been considered. Acoustic array logging tools have similarities with a seismic acquisition array, with an acoustic transmitter replacing the seismic explosive shot, air-gun, or vibroseis truck; and the receiver array of 8 or more elements a shorter equivalent of a geophone or hydrophone array. Reflections from fractures or bed boundaries with an acoustic impedance contrast in the vicinity of the wellbore follow the same basic seismic principles (figure 3). Reflection amplitudes are small due to attenuation experienced in the longer travel path through the near well bore formations, so signal enhancement is required. Similar signal processing techniques to those used in seismic processing are utilized, however additional steps are required to separate reflected data from direct arrivals and up-going and down-going wave energy (Tang et al., 2007). Previous near-borehole imaging methods used monopole P-wave data, which does not contain

any information about reflector azimuth as monopole sources are omnidirectional. It was determined that directional acoustic measurement using dipole tools has the potential to measure reflector azimuth (Tang, 2004). The lower frequency of dipole shear waves (2 to 3 kHz) tend to penetrate deeper than monopole waves due to the low-pass quality of rocks, enabling a stronger reflector signal.

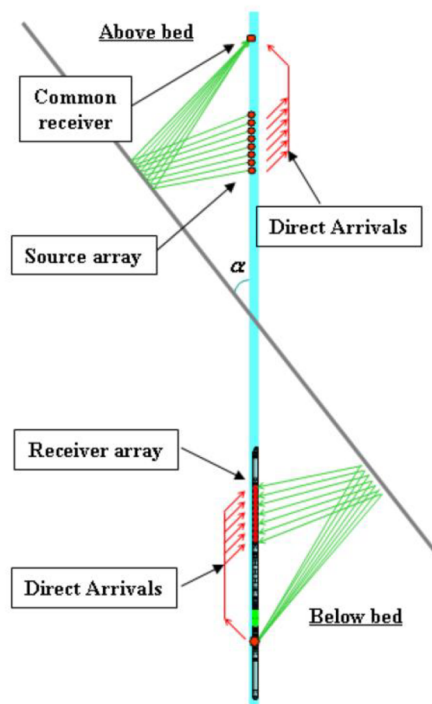


Figure 3—An acoustic tool positioned above or below a reflective surface, illustrating the up-going and down-going travel paths of reflective energy used for deep shear wave imaging, and direct arrivals as used for most other borehole acoustic applications.

Cross dipole waveforms and orientation data required for deep shear wave imaging (DSWI) processing. Data could be available for wells logged before this processing was developed, as the acoustic logging tools were already in service and recording the required acquisition modes. The depth of investigation is dependent on formation velocities and the length of recorded waveform. Normal cross-dipole waveform acquisition time (as expected for historical data) and downhole velocities allow DOI of approximately 50ft. Extended time acquisition is available for newly logged wells to increase the DOI of this method to 100ft or beyond.

Reflective surfaces may intersect the borehole, in which case borehole resistivity or ultrasonic imaging logs can independently verify the existence on the reflector, and the extent of the surface away from the borehole may be determined from the DSWI. Reflectors that do not intersect the borehole can also be mapped using this imaging technique and may provide information about the existing fracture network to aid in reservoir drainage, or to identify possible paths for unexpected water breakthrough. The distance to nearby bed boundaries can be mapped, for accurate determination of drainage volume. Cased hole applications are possible, pending a measurable deviation for high side alignment and acceptable wellbore quality.

Using an acoustic logging tool for reflector imaging has some advantages over surface seismic or vertical seismic profiles (VSP) when looking for small or localised formation changes. The higher frequency content of acoustic logging tools increases the resolution of reflectors, enabling the imaging of events with sub-seismic resolution. Unlike surface seismic or VSP, the travel paths do not need to travel from surface through overlying formations, which is particularly useful when a formation would create a blind spot on seismic data.

Conclusions

Historic seismic data surveys are commonly reprocessed to enhance data quality, or to utilize new processing techniques. The fullwave data from wireline or LWD acoustic tools, when available, offers similar opportunities to reprocess and extract additional information to aid in the ongoing understanding of an oilfield. Appropriate fullwave acoustic data from the historic well must have been recorded and stored for the intervening period. Challenges in retrieving data including outdated data formats, old storage media and hardware availability, and media degradation may all limit the opportunity to revisit these datasets, and serve as a reminder to maintain robust archives of important data to ensure future data access. Many processing options are available to fill knowledge gaps or to constrain newly discovered reservoir challenges, including but not limited to basic slowness reprocessing, resolution enhancement, permeability estimation, anisotropy determination, and deep shear wave imaging. In particular, the ability of deep shear wave imaging to move away from the immediate wellbore vicinity usually measured by acoustic tools and locate reflective surfaces many tens of feet away from the borehole demonstrates the possibilities available in old acoustic datasets. As yet undeveloped processes or techniques may become available in future which could help to solve reservoir challenges that are not currently realised.

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